

## Git Basics

1. Initialize a Repository

```
git init
```

2. Clone a Repository

```
git clone <repository-url>
```

3. Check Repository Status

```
git status
```

4. Add Files to Staging Area

```
git add <file-name> # Add specific file
```

```
git add . # Add all files
```

5. Commit Changes

```
git commit -m "Commit message"
```

6. Push Changes to Remote

```
git push origin <branch-name>
```

7. Pull Changes from Remote

```
git pull origin <branch-name>
```

8. View Commit History

```
git log
```

9. Show Changes (Unstaged)

```
git diff
```

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## Branching

1. Create a New Branch

```
git branch <branch-name>
```

2. Switch to a Branch

```
git checkout <branch-name>
```

3. Create and Switch to a New Branch

```
git checkout -b <branch-name>
```

4. List All Branches

```
git branch
```

5. Delete a Branch

```
git branch -d <branch-name>
```

6. Merge Branches

```
git merge <branch-name>
```

7. Rebase Current Branch

```
git rebase <branch-name>
```

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## Remote Repositories

1. **Add a Remote Repository**

```
git remote add origin <repository-url>
```

2. **View Remote Repositories**

```
git remote -v
```

3. **Remove a Remote Repository**

```
git remote remove origin
```

4. **Fetch Changes from Remote**

```
git fetch
```

5. **Push Tags to Remote**

```
git push --tags
```

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## Undoing Changes

1. **Unstage a File**

```
git reset <file-name>
```

2. **Revert Changes in a File**

```
git checkout -- <file-name>
```

3. **Reset to a Previous Commit**

```
git reset --hard <commit-hash>
```

4. **Revert a Commit**

```
git revert <commit-hash>
```

5. **Stash Changes**

```
git stash
```

6. **Apply Stashed Changes**

```
git stash apply
```

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## Tags

1. **Create a Tag**

```
git tag <tag-name>
```

2. **List Tags**

```
git tag
```

3. **Delete a Tag**

```
git tag -d <tag-name>
```

4. **Push Tags to Remote**

```
git push --tags
```

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## File and Directory Operations

### 1. List Files and Directories

```
ls
```

### 2. Change Directory

```
cd <directory-name>
```

### 3. Create a New Directory

```
mkdir <directory-name>
```

### 4. Remove a File

```
rm <file-name>
```

### 5. Remove a Directory

```
rm -r <directory-name>
```

### 6. Move or Rename a File

```
mv <old-name> <new-name>
```

### 7. Copy a File

```
cp <source> <destination>
```

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## Miscellaneous

### 1. Clear the Terminal

```
clear
```

### 2. Check Current Directory

```
pwd
```

### 3. Open Current Directory in File Explorer

```
explorer .
```

### 4. Check Git Version

```
git --version
```

### 5. View Git Configuration

```
git config --list
```

### 6. Set Global Username/Email

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your.email@example.com"
```